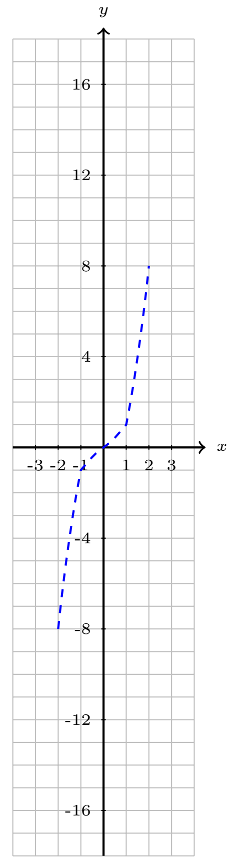
Unit 6: Cubic & Cube Root Functions Name: ______________________
Dilations & Reflections Practice Period: ________

Process: (1) Find a (2) Multiply each y-value by a (3) Plot new points (4) Draw the curve

1. $f(x) = 2x^3$ *Cubic — Vertical Stretch*



$a =$ _____

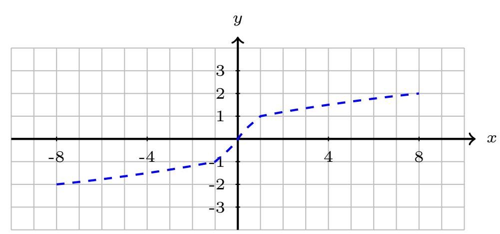
Effect: Stretch / Compress / Reflect

Parent	$y \times a$	**New**
$(-2, -8)$		(,)
$(-1, -1)$		(,)
$(0, 0)$		(,)
$(1, 1)$		(,)
$(2, 8)$		(,)

2. $f(x) = -\sqrt[3]{x}$ *Cube Root — Reflection*

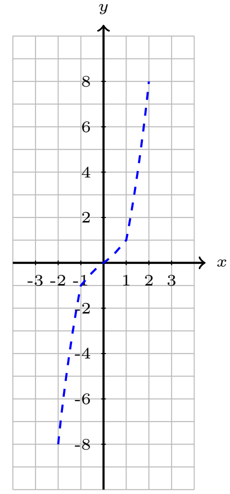
$a =$ _____

Effect: Stretch / Compress / Reflect



Parent	$y \times a$	**New**
$(-8, -2)$		(,)
$(-1, -1)$		(,)
$(0, 0)$		(,)
$(1, 1)$		(,)
$(8, 2)$		(,)

3. $f(x) = \frac{1}{2}x^3$ *Cubic — Vertical Compression*



$a =$ ______

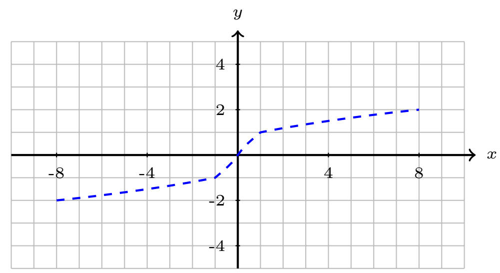
Effect: Stretch / Compress / Reflect

Parent	$y \times a$	**New**
$(-2, -8)$		(,)
$(-1, -1)$		(,)
$(0, 0)$		(,)
$(1, 1)$		(,)
$(2, 8)$		(,)

4. $f(x) = -2\sqrt[3]{x}$ *Cube Root — Stretch + Reflection*

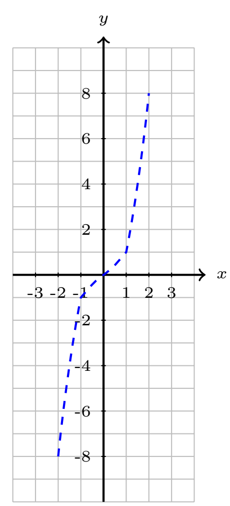
$a =$ ______

Effect: Stretch / Compress / Reflect



Parent	$y \times a$	**New**
$(-8, -2)$		(,)
$(-1, -1)$		(,)
$(0, 0)$		(,)
$(1, 1)$		(,)
$(8, 2)$		(,)

5. $f(x) = -x^3$ *Cubic — Reflection only*



$a =$ ______

Effect: Stretch / Compress / Reflect

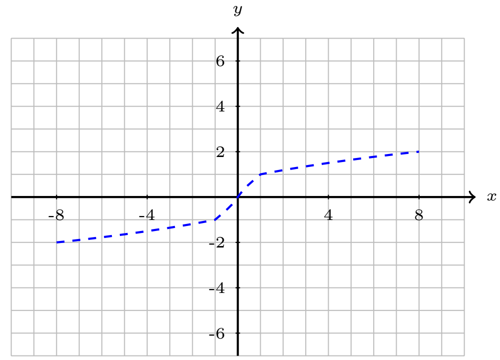
Parent	$y \times a$	**New**
$(-2, -8)$		(,)
$(-1, -1)$		(,)
$(0, 0)$		(,)
$(1, 1)$		(,)
$(2, 8)$		(,)

6. $f(x) = 3\sqrt[3]{x}$ *Cube Root — Vertical Stretch*

$a =$ ______

Effect: Stretch / Compress / Reflect

Parent	$y \times a$	**New**
$(-8, -2)$		(,)
$(-1, -1)$		(,)
$(0, 0)$		(,)
$(1, 1)$		(,)
$(8, 2)$		(,)

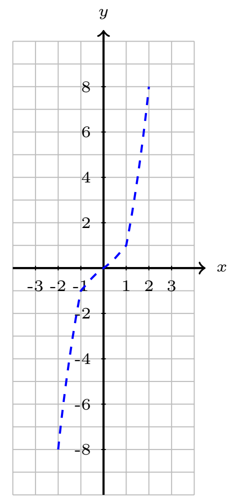


7. $f(x) = -\frac{1}{2}x^3$ *Cubic — Compression + Reflection*

$a =$ ______

Effect: Stretch / Compress / Reflect

Parent	$y \times a$	**New**
$(-2, -8)$		(,)
$(-1, -1)$		(,)
$(0, 0)$		(,)
$(1, 1)$		(,)
$(2, 8)$		(,)

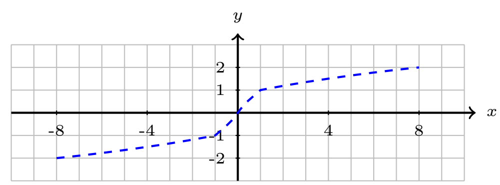


8. $f(x) = \frac{1}{3}\sqrt[3]{x}$ *Cube Root — Vertical Compression*

$a =$ ______

Effect: Stretch / Compress / Reflect

Parent	$y \times a$	**New**
$(-8, -2)$		(,)
$(-1, -1)$		(,)
$(0, 0)$		(,)
$(1, 1)$		(,)
$(8, 2)$		(,)



Summary Questions:

9. Without graphing, describe the difference between $f(x) = 3x^3$ and $f(x) = \frac{1}{3}x^3$.

10. A student says $f(x) = -2x^3$ is "flipped and shorter." Correct their mistake.

11. If $a = -\frac{1}{4}$, describe ALL the transformations that happen to the parent function.